EMG4 post-hoc neutral segment analysis

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Overview

This document contains the preparation of the EMG4 data from the neutral segment for analysis. As well as the post-hoc analysis of the neutral segment in between the state adjective and affect reason segments.

```
library(readxl)
library(sjPlot)
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)
```

Load data from all 3 segments

1. Preprocessing Neutral segment

Preprocessing is needed to transform the data from wide to long format based on the 100ms bins for the corrugator response. The relevant variables are computed for the different conditions. Another MoralityValence variable is computed to estimate the final no-intercept model. Furthermore, we've computed linear, quadratic and cubic time trends for each condition (moral, immoral).

2. MLM Neutral segment

Building the models

Because our main focus was on the development of the corrugator response over time, we used a growth curve model approach (Mirman, 2015; Peck and Devore, 2008) with specific analysis designs that were optimized for assessing and comparing time trends across conditions. We first modelled participants and items as random factors. To assess the effect of our manipulations on the average activation across an entire segment, we first added morality dimension (moral vs. immoral) as a fixed factor in the model for the character manipulation segment, and morality dimension (moral vs. immoral), critical event valence (positive vs. negative) and interaction as fixed factors in the model for the affective state adjective and affect reason segments. Afterwards the interaction was added to the random part of the model as a random slope.

Next, linear, quadratic, and cubic trends will be added as covariates in the fixed part of the model, and if significantly improving the model also to the random part, see below. Time trend (e.g. linear) components will be added per condition to maintain flexibility in building the model, and to avoid forcing the model to fit, for example, a linear trend for all conditions when only one condition contains a significant linear component. All trend components will be centered to avoid correlation between trends (fixed effects final model intercepts therefore reflect the average corrugator activity across the entire segment, not the level at which corrugator activity intercepts the y-axis). By using trends up to the cubic component, we achieve some flexibility to fit responses without over-fitting or losing explanatory power (Mirman, 2015). Because we are particularly interested in temporal developments, we will add random slopes for subjects for each time trend that initially improves the model (as well as standard random intercepts for subject and item).

variables:

dependent variable: Neutral_response random factors: sjCode, Markertrial
independent variables: Morality (102=moral, 103=immoral), StateAdjectiveCondition (104=positive, 105=negative), t1_ (continuous linear trend per condition, will be centered for easier interpretation of the estimates), t2_ (continuous quadratic trend per condition, will be centered for easier interpretation of the estimates) , t3_ (continuous cubic trend per condition, will be centered for easier interpretation of the estimates)

Summary of best Model NeutralM13

The best model is Model 13: Neutral_response ~ MoralityCondition + StateAdjectiveCondition + MoralityCondition:StateAdjectiveCondition + t1_immoral_pos + t2_immoral_pos + (1 + t1_immoral_pos | sjCode) + (1 | Markertrial). This is the summary of that model.

```
load(paste0(getwd(),"/WORCS_EMG4/results/EMG4_Neutral_final_model_60sjs.RData"))
tab_model(NeutralM13)
```

	Neutral Segment (M13, *df* = 93484)		
Predictors	*Estimates (95% CI)*	*z*	*p*
(Intercept)	108.60 (98.31, 118.89)	20.68	<0.001*
MoralityCondition [Immoral]	7.89 (5.84, 9.94)	7.54	<0.001*
StateAdjectiveCondition [Negative]	19.86 (18.25, 21.47)	24.23	<0.001*
t1 immoral positive	-3.39 (-7.87, 1.10)	-1.48	0.14
t2 immoral positive	2.66	2.13	0.03*

	(0.21, 5.10)		
MoralityCondition [Immoral] * StateAdjectiveCondition [Negative]	-10.40 (-13.00, -7.80)	-7.84	<0.001*
Random Effects			
σ^2	7824.57		
τ_{00} Markertrial	124.65		
τ_{00} sjCode	1516.98		
τ_{11} sjCode.t1_immoral_positive	275.49		
ρ_{01} sjCode.t1_immoral_positive	-0.86		
ICC	0.18		
Deviance	1104178.80		
log-Likelihood	-552089.40		

Post-hoc analyses

```
NeutralM13_emmeans_contrasts = emmeans(NeutralM13,specs = tukey ~ MoralityCondition * StateA
djectiveCondition, lmer.df = "satterthwaite", lmerTest.limit=93500, simple = "each", combine
= T, adjust = "sidak", reverse = T)
NeutralM13_emmeans_contrasts$emmeans

##  MoralityCondition StateAdjectiveCondition emmean   SE   df lower.CL upper.CL
##  Moral             Positive                   109 5.25 70.4     98.5      119
##  Immoral           Positive                   117 5.27 71.5    106.3      127
##  Moral             Negative                   129 5.25 70.4    118.3      139
##  Immoral           Negative                   126 5.25 70.4    115.8      137
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95

NeutralM13_emmeans_contrasts$contrasts %>%
     summary(infer = TRUE)

##  StateAdjectiveCondition MoralityCondition contrast            estimate    SE
##  Positive                .                 Immoral - Moral         7.89 1.046
##  Negative                .                 Immoral - Moral        -2.51 0.817
##  .                       Moral             Negative - Positive    19.86 0.820
##  .                       Immoral           Negative - Positive     9.46 1.045
##     df lower.CL upper.CL t.ratio p.value
##  93322     5.28   10.494   7.545  <.0001
##  93322    -4.55   -0.476  -3.074  0.0084
##  93334    17.82   21.902  24.230  <.0001
##  93327     6.86   12.061   9.056  <.0001
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: sidak method for 4 tests
```

Plot of regression lines for the different State Adjectives for the Morality manipulation in the Affect Reason segment:

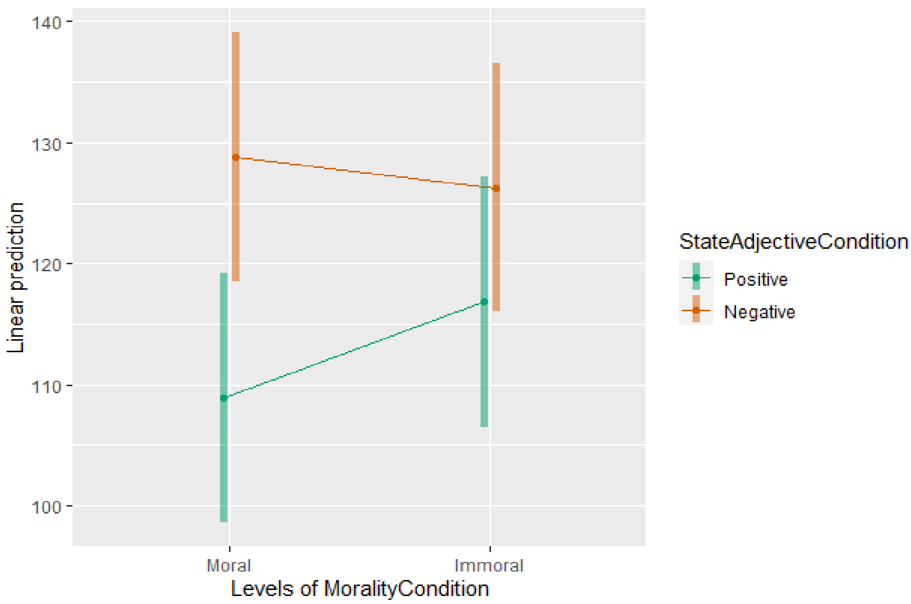
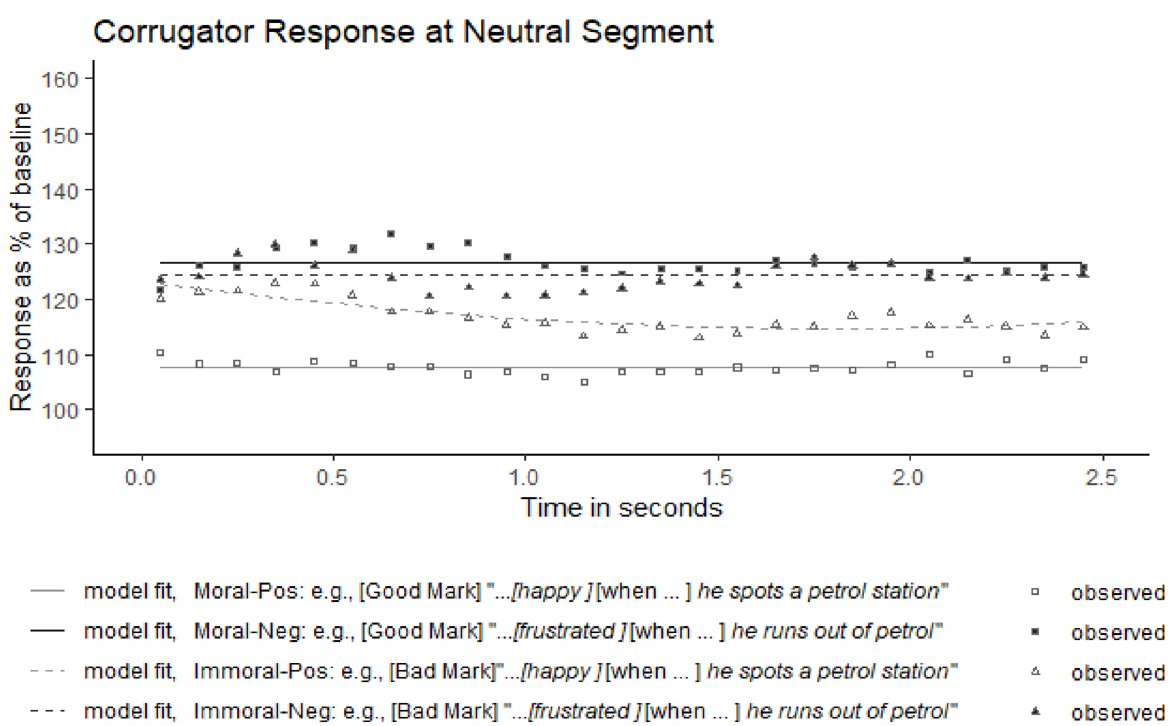


Figure on neutral segment with CI

The comparison between t1_immoral_pos and t2_immoral_pos is redundant since they refer to different time trends. However, the negative t1 component indicates a downward slope with a rise-fall pattern, as indicated by the positive t2 component.

3. Figures

Neutral segment



Adjective + neutral + reason